

Multiple Account Objects in a List

Representing each account in a separate global variable works, but this is not a good approach when we need to handle a large number of objects. A bank would need a way to handle an arbitrary number of accounts.

Whenever we need an arbitrary number of pieces of data, a list is the typical solution.

In this version of the test code, we'll start with an empty list of `Account` objects. Every time a user opens an account, we'll instantiate an `Account` object and append the resulting object onto our list. The account number for any given account will be the index of the account in the list, starting with 0. Again, we'll start by creating a test account for Joe and one for Mary, as shown in [Listing 4-4](#).

File: BankOOP2_ListOfAccountObjects/Main_Bank_Version2.py

```
# Test program using accounts
# Version 2, using a list of accounts

# Bring in all the code from the Account class file
from Account import *

# Start off with an empty list of accounts
accountsList = [ ] ❶

# Create two accounts
oAccount = Account('Joe', 100, 'JoesPassword') ❷
accountsList.append(oAccount)
print("Joe's account number is 0")

oAccount = Account('Mary', 12345, 'MarysPassword') ❸
accountsList.append(oAccount)
print("Mary's account number is 1")

accountsList[0].show() ❹
accountsList[1].show()
print()

# Call some methods on the different accounts
print('Calling methods of the two accounts ...')
accountsList[0].deposit(50, 'JoesPassword') ❺
```